

Fahmida Yasmin

Computer Science Engineer — AI/ML Researcher — Full-Stack Developer

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About Me

I'm a Computer Science Engineer with a strong academic record and a focus on Machine Learning and data-driven decision-making. My work bridges the gap between academic research and scalable software engineering, with experience in geospatial analytics, blockchain technology, and full-stack development. I am driven by the challenge of applying AI to solve complex urban and clinical problems, ensuring that my research results are not just theoretical but practically applicable.

Work Experience

International Islamic University Chittagong (IIUC)

Teaching Assistant, Department of Computer Science and Engineering

Aug 2026 – Jan 2027

Chattogram, Bangladesh

FlyRank AI

Machine Learning Intern

Jul 2026 – Sep 2026

Remote

Education

B.Sc. in Computer Science Engineering

International Islamic University Chittagong — Current CGPA: 3.91

2022 – Present

Higher Secondary Certificate (HSC)

Cambrian College

2019 – 2021

Secondary School Certificate (SSC)

Silver Bells Girls High School

2007 – 2019

Achievements

Best Female Team Award — Project Showcase, IIUC CSE Fest 2026

International Islamic University Chittagong

Jun 2026

Publications

Crime Hotspot-Driven Safe Navigation and IoT Emergency Alert System for Women's Safety in Bangladesh

Undergraduate Thesis, International Islamic University Chittagong

Ongoing

Designing a spatio-temporal crime risk scoring model and a modified safety-aware pathfinding algorithm, paired with an IoT-based emergency alert module for real-time distress response, to route and protect women away from high-risk areas.

UrbaSafe: A Saliency-Guided Closed-Loop Framework for Urban Safety Perception and Generative Renewal

Conference Paper

Under Review

Proposed a dual-model framework combining a ConvNeXt-Tiny perceptual safety scorer with a Grad-CAM-guided saliency-to-mask pipeline and Stable Diffusion Inpainting, enabling closed-loop detection and AI-generated renewal of visually unsafe urban streetscapes.

A Comparative Study on Cardiovascular Disease Prediction via Machine Learning and SHAP Interpretability Analysis 2026

Conference Paper, Under Review

Applied five ML classifiers with SHAP explainability for accessible, interpretable cardiovascular disease prediction.

An Efficient NFC Blockchain Model for Detecting Counterfeit and Expired Products 2025

Conference Paper, Taylor & Francis

Proposed an NFC and blockchain-based framework integrated with IoT to detect and prevent counterfeit and expired products.

Projects

IIUC Nexus — University Community Super-App

Tech Stack: Flutter · Laravel · Appwrite Cloud · Android/iOS/Web

- Built the official university community super-app of IIUC, digitally uniting students, faculty, and campus clubs on a single trusted, bilingual (English/Bengali) platform used across Android, iOS, and Web.
- Implemented a secure dual-backend (Laravel + Appwrite Cloud) with a serverless verification system that automatically rejects non-IIUC emails, ensuring only verified IIUC members can register and join.
- Integrated campus community engagement, academic resources, faculty consultation, and an AI university assistant into one unified cross-platform app (v1.0.0).

Women's Safety Navigation System

Tech Stack: MongoDB · Express · React · Node.js · OpenStreetMap · Python · IoT · matplotlib · seaborn

- Building a full-stack web platform visualizing spatio-temporal crime hotspot data on interactive OpenStreetMap maps to support safety-optimized navigation for women in Bangladesh.
- Integrating real-time IoT physiological sensor data with a backend GSM emergency alert system for live distress response.

Blockchain-Based Decentralized NFC Product Authentication System

Tech Stack: Ethereum · Solidity · Ganache · Truffle · MetaMask · Web3.js · HTML · CSS · JavaScript · Bootstrap

- Developed a decentralized platform featuring role-based access control and an immutable ledger.
- Implemented consensus-based transactions and real-time product verification with expiry alerts.
- Built transparent scan history to enhance product traceability across the supply chain.

Automated Traffic Systemizer

Tech Stack: Arduino UNO · C++ · Servo Motor · Ultrasonic Sensor (HC-SR04) · 7-Segment Display · LED Array

- Built a physical smart traffic control system integrating a servo-controlled barrier, sensor-driven dynamic street lighting, and 7-segment countdown display on real hardware.
- Programmed proximity-based LED sequencing using ultrasonic distance sensing for energy-efficient road illumination.
- Demonstrated end-to-end embedded IoT logic: sensor input, real-time actuator control, and synchronized multi-component coordination.

Event Management System

Tech Stack: HTML · CSS · Bootstrap · PHP · MySQL · JavaScript · Ajax

- Built a responsive web application with MySQL database backend enabling event booking and vendor selection.
- Implemented automated email confirmations and secure feedback from verified users.
- Enabled real-time feedback updates using dynamic frontend interactions with Ajax.

Skills

Languages: Python, C++, JavaScript, Solidity, PHP

ML/Data: Pandas, Scikit-learn, SHAP, Matplotlib, Seaborn, Geopandas

Web/Full-Stack: React, Node.js, Express, MongoDB, MySQL, HTML, CSS, Bootstrap, Django, Flutter, Laravel

Blockchain/IoT: Ethereum, Web3.js, Truffle, Ganache, Arduino, Embedded Systems